

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX X-Axis Glass Scale Replacement Procedure	Effective Date	25 Jan 2017

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	25 Jan 2017	First Release	TAN LIANG KAI

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A. X-Axis Glass Scale

I. X-Axis glass scale removal

S2EX machine X-Axis glass scale assembly Part ID: 4XASP-A144

1. Turn off x-ray source & open front access door.
2. Push X-Axis stage to right hand side and loosen 2 screws as shown in Figure 1. Remove X-Axis encoder.

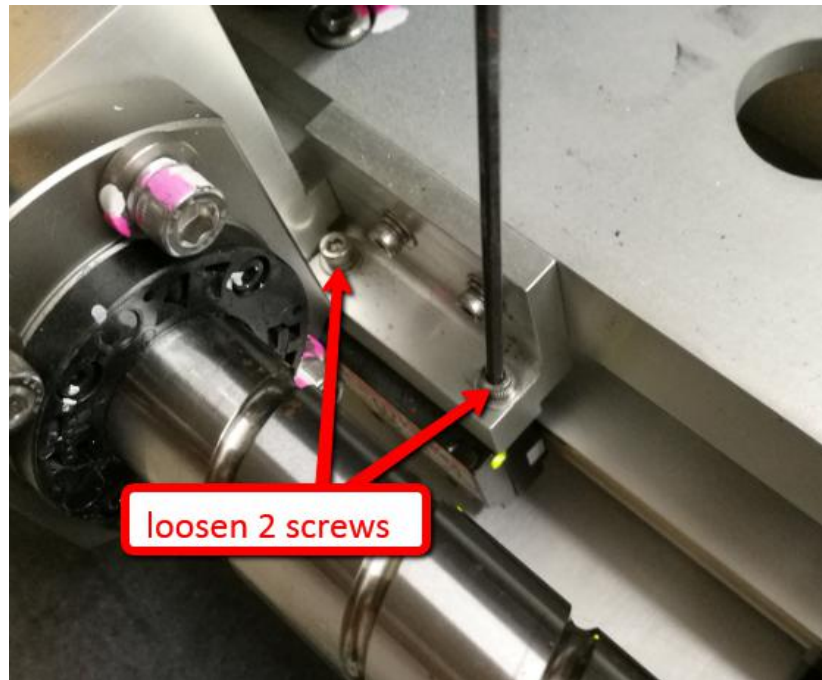


Figure 1 X-axis encoder removal

3. Remove all spring clamps, home sensor and limit sensor as shown in Figure 2.

Remark: Perform minor marking on current limit sensor & home sensor position for ease of adjustment.

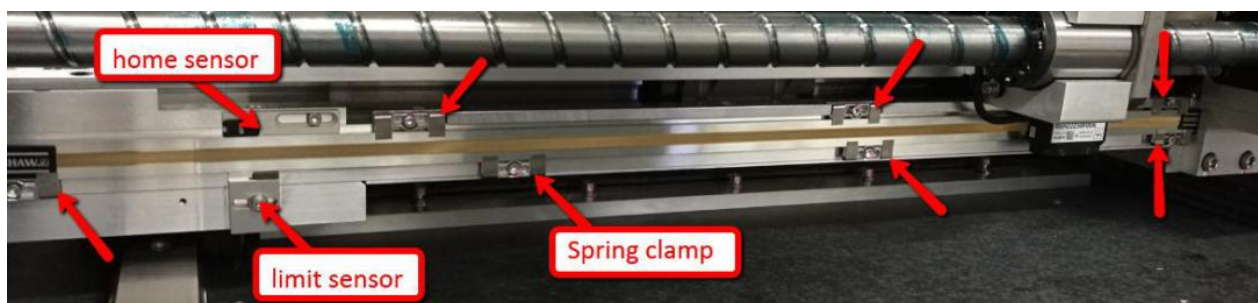


Figure 2 Glass scale

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4. Cut RTV glue which secure glass scale as shown in Figure 3.

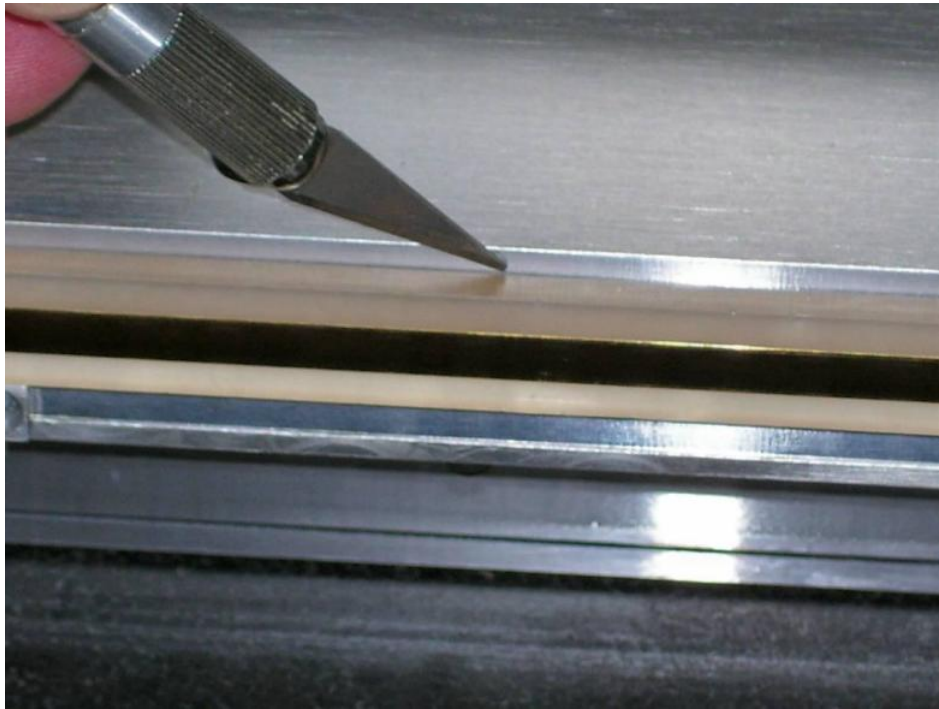


Figure 3 RTV glue removal

5. Gentle slide out glass scale towards left direction as shown in Figure 4.

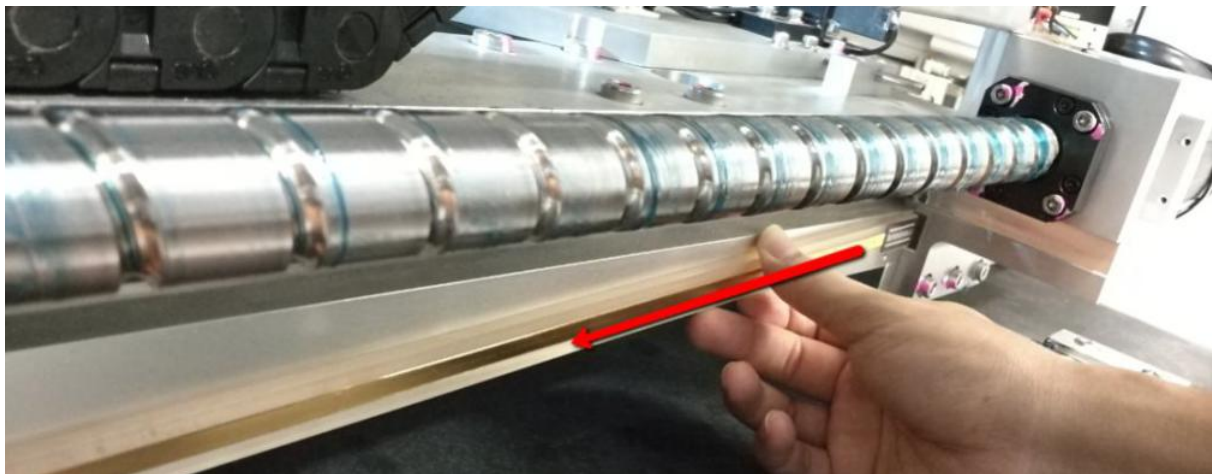


Figure 4 Glass scale removal

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II. Replacement of the X-Axis Glass Scale

1. Insert X-Axis glass scale into slot as shown in Figure 5.

Remark: Caution on glass scale orientation.

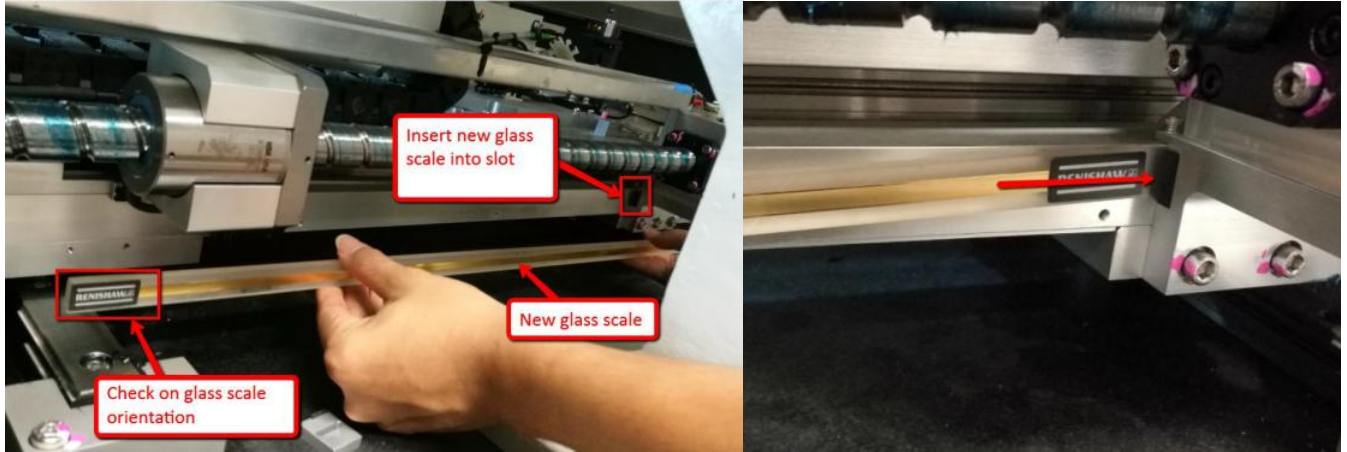


Figure 5 Glass scale assembly

2. Position glass scale lining up at termination tag of the glass scale with spring clamp.



Figure 6 Glass scale lining up (Reference only)

3. Assemble Home Sensor & Limit Sensor by refer to origin location (By refer to Step 3). Tighten all screws which secure spring clamps as shown in Figure 7.

Remark: Make sure spring clamps are well-positioned and do not contain any friction with encoder while moving X-axis stage.

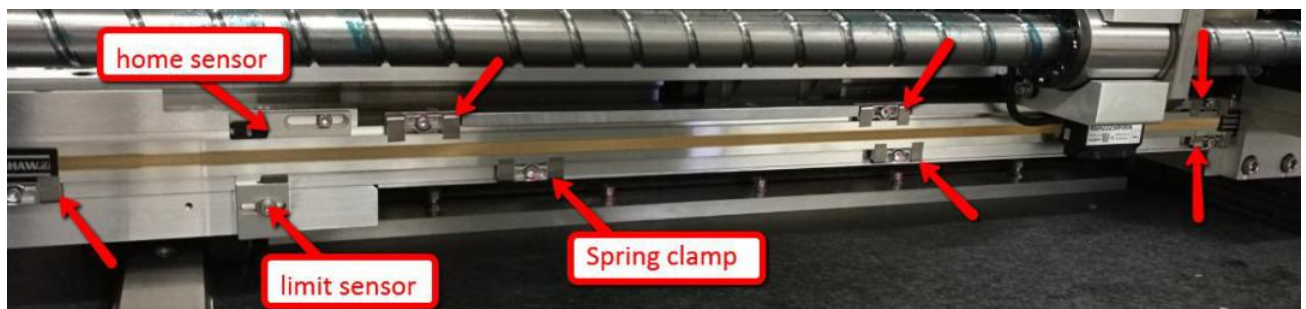


Figure 7 Glass scale replacement

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4. Assemble X-axis encoder & tighten 2 screws as shown in Figure 8..

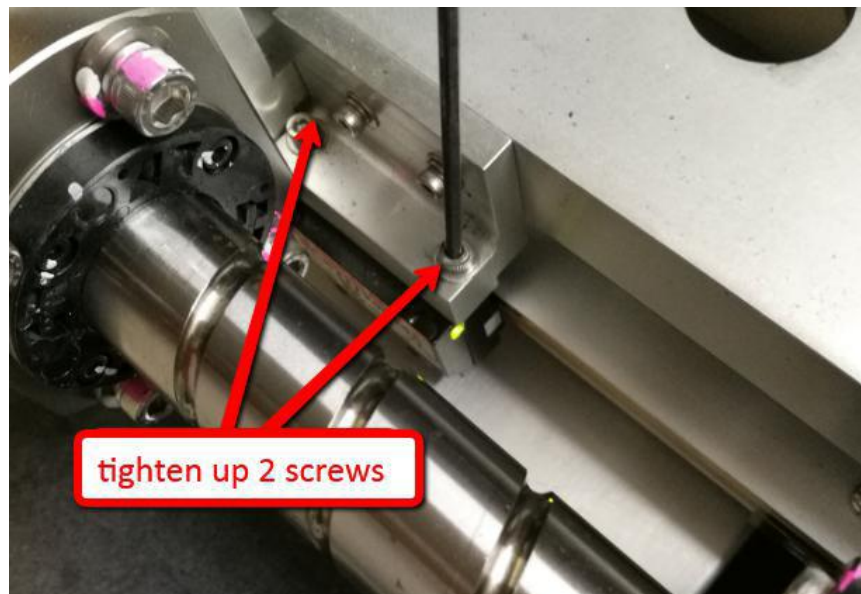


Figure 8 X-Axis encoder replacement

5. Move X-Axis stage to minimum hard stop (left side) and maximum hard stop (right side). Visual check X-Axis encoder which require green light illuminated along the way as shown in Figure 9. If adjustment require, may refer to **“AXI V810 X-Axis Read Head Encoder Replacement Procedure”**.

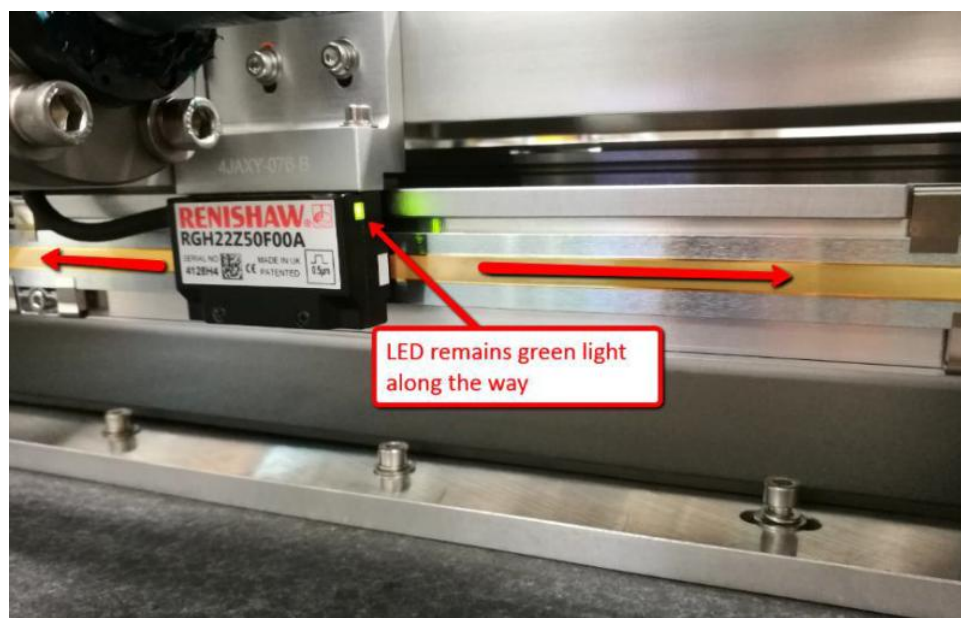


Figure 9 Encoder checking

6. Close front access door.

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7. Open V810 > Services > Subsystem Status & Control > Panel Positioner > Initialize

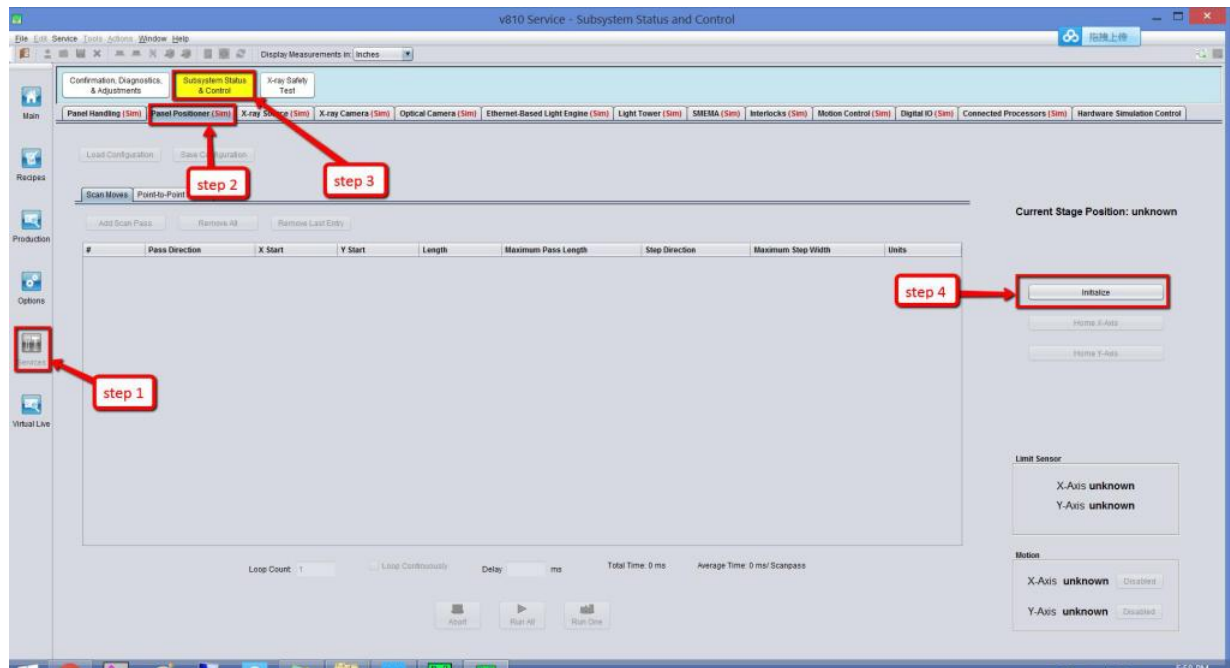


Figure 10 Initialize Panel Positioner

8. Open front access door.

9. Disable X-Axis motion at Panel Positioner and perform below checking.

- a) Do not move XY Stage and may check Limit sensor status at Panel Positioner where must stated as "Clear" during home position as shown in Figure 11.

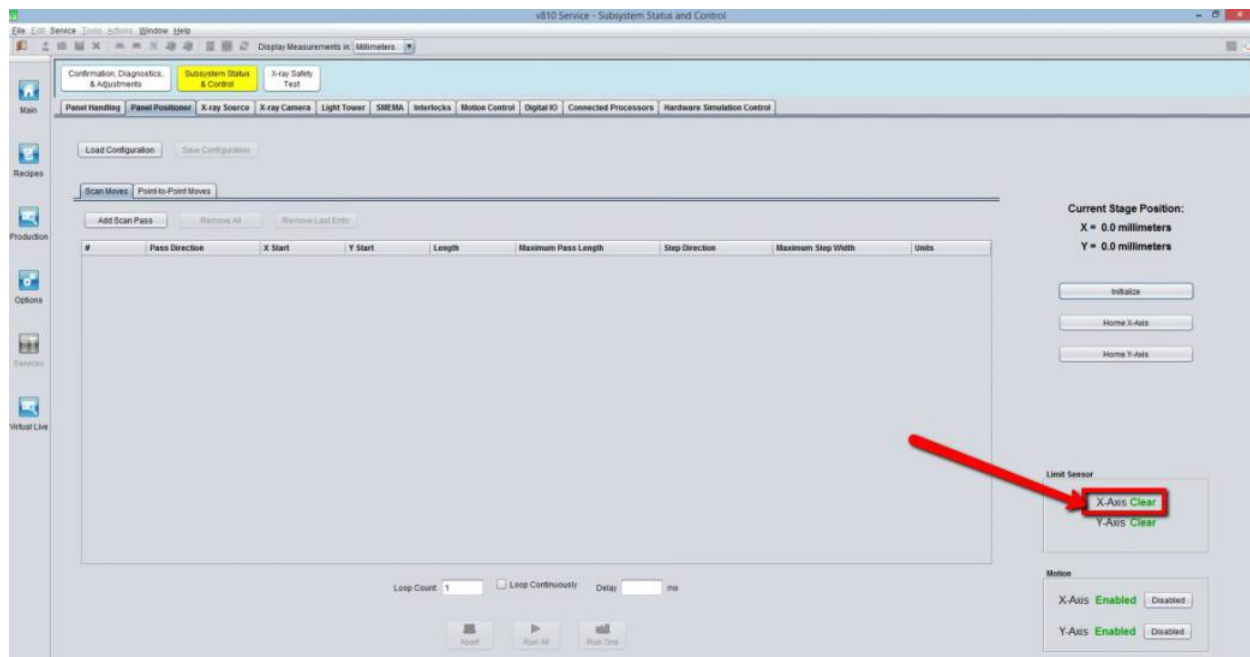


Figure 11 X axis Limit Sensor (CLEAR)

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- b) Move X axis stage towards minimum hard stop (Left Side) & check Limit sensor where must stated as “ Blocked” as shown in Figure 12.

Remark: If result is not meet to Step 9a ~9b requirement ,may refer to “AXI V810 X-Axis Limit Sensor Adjustment Procedure” for further adjustment.

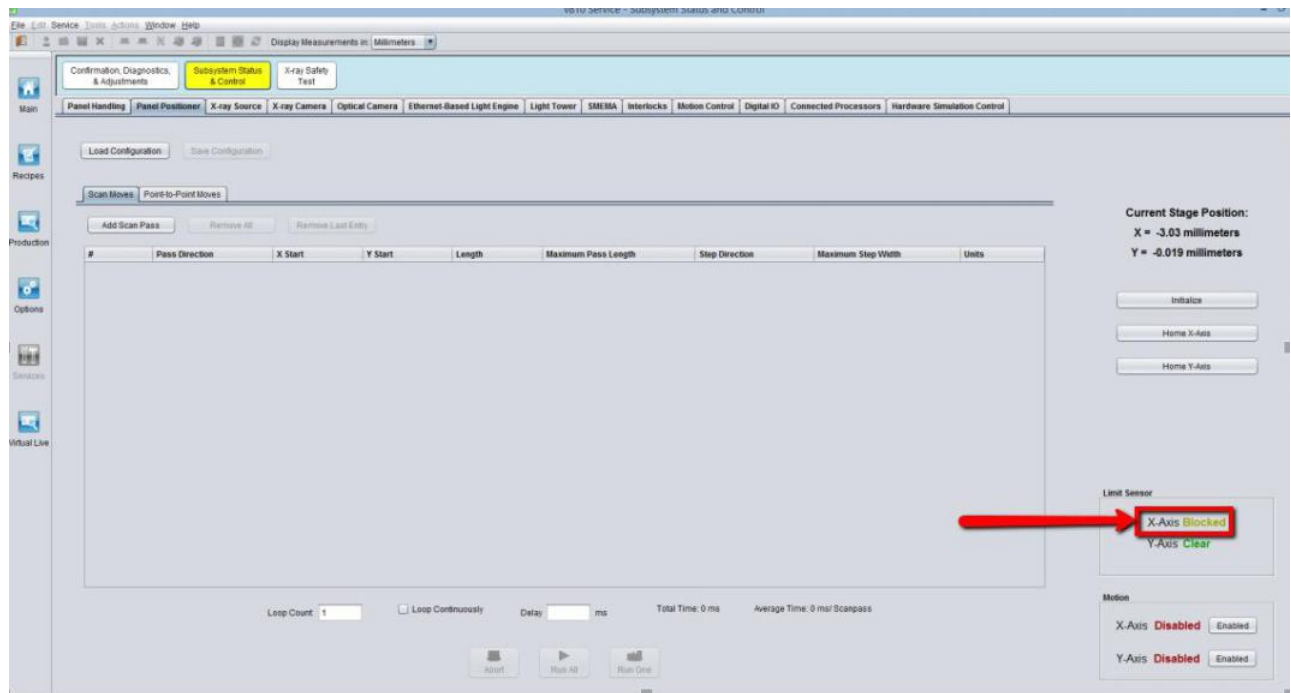


Figure 12 X axis Limit Sensor (BLOCKED)

- c) Move X axis stage towards minimum hard stop(Left side).
d) Check the value for X-axis in current stage position must within 3.00 ~ 3.30 mm.

Remark: If result is not meet to Step 9c ~9d requirement ,may refer to “AXI V810 X-Axis Home Sensor Adjustment Procedure” for further adjustment.



Figure 13 X axis (Current Stage Position)

10. Run 3 loops of long scan and 1 loop of CD &A.